KIIT Deemed to be University
Online Mid Semester Examination(Spring Semester-2021)

Subject Name & Code: Web Technology **Applicable to Courses: CSE/IT/CSSE (BTech)**

Full Marks=20 **Time:1 Hour**

SECTION-A(Answer All Questions. All questions carry 2 Marks)

Time:20 Minutes **(5×2=10 Marks)**

Question No	Question Type (MCQ/SAT)	Question	Answer Key(if MCQ)	CO Mapping

Q.No:1(a)		What is the result of following piece of code?
		public class Person{
		public void talk(){
		System.out.print("
		}
		}
		public class Student extends Perso
		public void talk(){
		System.out.print("
		}
		}
		public class Test{
		public static void main(St
		Person p = new Stu
		p.talk();
		}
		}
		a) I am a Person
		b) I am a Student
		c) I am a Person I am a Student
		d) I am a student I am a person
			B	

		Write the output?
		class A {
		A() {}
		A f1() { return this;}
		void show() { System.out.println(" A class"); }
		public static void main(String x[]) {
		new A(). f1().show(); }
		}
		A. compilation error
			C	

		B. run time error C. A class Displayed nothing		

		public class Final
		{
		final int assign = 30;
		
		public static void main(String[] args)
		{
		final int assign;
		
		Final f = new Final();
		assign = 20;
		
		System.out.println(assign);
		System.out.println(f.assign);
		
		System.out.println(f.process(f.assign));
		}
		int process(int a)
		{
		return a + 5;
		}
		}
		
		A. 20
		20
		25
		
		B. 20
		30
		35
		
		C. Compilation Error
		
		D. Runtime Error
			B	

		FFind the output of the following program:
		
		class Demo {
		public static void main(String[] args) {
		
		int a = -5;
		int b = 8;
		for (int i = 0; i < 8; i++)
		{
		if((++a > 2) || (b-- > 2))
		{
		a++;
		}
		}
		System.out.println("a="+ a + " " + "b="+b);
		
		}
		}
		A. a=-9 b=4
		B. a=11 b=4
		C. a=10 b=3
		B. a=9 b=3
			B	

Q.No:1(b)		**1. Find output of the following program**	D	

		public class Test
		{
		public static void main(String[] args) {
		for(int i = 0; i < 3; i++){
		for(char c = 'A' ; c < 'D'; c++)
		if(i % 2 == 0){
		System.out.print(c);
		break;
		}
		System.out.print(i);
		}
		
		}
		}
		
		A. A012A
		
		B. A0A1A2
		C. AAA012
		D. A01A2		

		Consider the following code snippet in Java.
		
		int a = -035;
		int b = a << 2;
		int c = a >> 2;
		System.out.println("b = " + b + ", c = " + c);
		
		The output of the above code snippet would be:
		(a) b = -140, c = -9
		(b) b = -140, c = -8
		(c) b = -116, c = -9
		(d) b = -116, c = -8
			D	

		Choose the most appropriate option for the following statement. Java compiler creates a constructor when - a) Constructor with no parameter is not defined in the class b) No constructor is defined in the class. c) main method is defined in the class D)None of the above	B	
		Choose the most appropriate option for the following statement. Polymorphism is achieved in Java by using - a) Method overloading technique b) Method overriding technique c) Inheritance D)All of the above	D	

Q.No:1(c)		What will be the correct line in the dotted part of main method to print the value of i to output?
		class example
		{
		static int i=10;
			D	

		public static void show()
		{
		System.out.print(i);
		}
		}
		class example1 {
		public static void main(String[] args)
		{
		example e = new example();
		…….
		}
		}
		a) e.show();
		b) System.out.print(this.i);
		b) example1.show() d) example.show();		

		Find the output of the following:
		public class Test
		{
		public static void main(String args[])
		{
		double[] mylist = {2,6,6,1,3,5};
		double max = mylist[0];
		int indexOfMax = 0;
		for(int i=1; i<mylist.length;i++)
		{
		if(mylist[i]>max)
		{
		max = mylist[i];
		indexOfMax = i;
		}
		}
		
		System.out.println(indexOfMax);
		}
		}
		
		a. 2
		b. 1
		c. 0
		5
			B	

		Which one of the following is a valid statement?
		a. char[] c=new char[];
		b. char[] c=new char[4];
		c. char[] c=new char();
			B	

		d. none of these		
		1. How many arguments can be passed to main()? a) Infinite b) Only 1 c) System Dependent d) None of the mentioned	A	

Q.No:1(d)		2. What will be the output of the following Java program, Command line execution is done as – "java Output This is a command Line"?
		
		class Output
		{
		public static void main(String args[])
		{
		
		System.out.print("args");
		}
		}
		
		a) This
		b) java Output This is a command Line
		c) This is a command Line
		d) Compilation Error
			C	

		What will be the output of the following Java program?
		
		class main_class
		{
		public static void main(String args[])
		{
		int x = 9;
		if (x == 9)
		{
		int x = 8;
		
		System.out.println(x);
		}
		}
		}
			C	

		a) 9 b) 8 c) Compilation error d) Runtime error		

		Write the output?
		class A {
		int x;
		A() {x=20 ; }
		A(int y){
		x=y;
		this();
		}
		public static void main(String z[]) {
		A p =A(10);
		System.out.println(p.x);
		}
		}
		A. Compilation error
		B. Run time error
		C. 10
		D. 20
			A	

		Which of the followings will lead to compilation error? i) float [] arr = new float(5) ii) float arr [] = {1.1f} iii) float [] arr = new float [5] iv) float arr [] = new float [] a) i AND iii b) i AND ii c) i AND iv d) ii AND iv	C	
Q.No:1(e)		The "Action" attribute in HTML forms specifies that a) Which HTTP method is used b) Which action is going on c) Where to submit form d) The auto completion of form	C	
		1. Which is the correct HTML for adding blue as a background colour? (i) <background>blue</background> (ii) <body bg="blue">> (iii) <body style="background-color:blue;">> (iv) <body	C	

		background-color="yellow; ">		
		Consider the HTML table definition given below: <table border=3> <tr> <td rowspan=2>ABC</td> <td colspan=2>DEF</td> </tr> <tr> <td>MNO</td> <td rowspan=2>PQR</td> </tr> <tr> <td colspan=2>XYZ</td> </tr> </table> How many number of rows in each column and the number of columns in each row present in the above table? a)(2,3,2) and (2,3,2) b(3,2,2) and (2,3,2) c)(2,3,3) and (2,3,3) d)(2,2,3) and (3,2,2)	A	
		In order to add space of 20px between cell content and its border of a table in html, which one of the following option is appropriate? a) th,td{ padding: 20px;} b) table{ margin:20px; } c) table{ border : 20px;} d) table{ padding: 20px; }	A	

SECTION-B(Answer Any One Question. Each Question carries 10 Marks)

Time: 30 Minutes **(1×10=10 Marks)**

Question No	Question	Evaluation scheme
Q.No:2	Create a class Data with data members: height and breadth of object given by the user and member functions get_data() to read the values and put_data() to display the values. Create another class Rectangle that inherits class Data and implement its methods areaRectangle () and perimeterRectangle() that computes the area and perimeter of a rectangle. Create another class Triangle that inherits class Data and implement its methods areaTriangle() and perimeterTriangle() that computes the area and perimeter of a triangle. [5] [b]Write a program in java to remove duplicate elements in an array. Eg: initial array: [2,3,8,8,6,5,8,9,2,5,5], the output will be a sorted array i.e: [2,3,5,6,8,9].What is dynamic method dispatch? Show with an example. [2+3]	1.5 marks each for creating the Rectangle and Triangle classes correctly , and 2 marks for integrating with Data class. 2 marks for correct program. 1 mark for definition and 2 marks for example.
Q.No:3	[a]Write a menu driven program in java which will display the following menu:- 1. Addition of two number 2. Subtraction of two number	1 mark each for the addition, subtraction and multiplication, and 2 marks for

<table>
<tr>
<td></td>
<td>3. Multiplication of two numbers
And the program will display the output accordingly. The program will accept the value of two numbers from the keyboard using Scanner class. The program will also ask at the end (Do you want to continue – Press – 1 for yes and 2 for no) [5]

[b] What are the features of Java which make it so popular ? Describe each feature. [5]</td>
<td>integrating with the output and Scanner class, and press 1/2.

1 mark for mentioning the features and 4 marks for description</td>
</tr>
<tr>
<td><u>Q.No:4</u></td>
<td>[a]Write a program to check user entered number is a neon number or peterson number or Buzz number using switch statement.Your program should validate all three types of numbers.
Neon Number: A positive number whose sum of digits of its square is equal to the original number itself. Example: 9^2=81 and 8+1=9 so 9 is a neon number
Peterson Number: A number is Peterson if the sum of factorials of each digit is equal to the sum of the number itself.
Example:
145 = !1 + !4 + !5==>1+4*3*2*1+5*4*3*2*1==>1+24+120==>**145**
Buzz Number: A number is said to be Buzz if it ends with 7 or is divisible by 7.
Example:147 is a Buzz number as it ends with 7 and also divisible by 7.
42 is a Buzz number because it is divisible by 7.
[5]
[b]While writing a program when does it become mandatory to use a default constructor? Explain with an example. Also justify how name conflicts are avoided using “this” keyword.
[1+2+2]</td>
<td>1 mark each for creation of Neon number/Peterson number/Buzz number, and 2 marks for correct integration and output using switch-case

1 mark for correct reason, 2 marks for example, and 2 marks for “this” keyword usage</td>
</tr>
<tr>
<td><u>Q.No:5</u></td>
<td>What is the difference between cell spacing and cell padding? What are tags and attributes in HTML? Name two tags which do not come in pairs. [2+2+1]

Design the following table using html [5]
<table>
<tr><td colspan="2">A</td><td>B</td></tr>
<tr><td rowspan="2">C</td><td>D</td><td>E</td></tr>
<tr><td colspan="2">F</td></tr>
</table></td>
<td>2 marks for difference. 2 marks for tags and attributes definition. 1 mark for naming the two tags

5 marks if all rowspan and colspan definitions correct, else part marking as deemed fit by</td>
</tr>
</table>

		faculty.
Q.No:6	[a] What are the advantages of using frames in HTML? [b]Write HTML code to display a web page as follows, and screen split to be done using frames: Google KIIT University IIT Kharagpur **This is the main page and content of the any link will be displayed here.**	3 marks for advantages of frames 7 marks to be awarded if all components correct and all hyperlinks and navigation working correctly. Else part marking as deemed fit by the evaluator
Q. No:7	Design an HTML form to select one student's roll number from a list of 10 roll numbers on the left hand side. On clicking the roll number, it will display the name, address and marks of five subjects in a tabular format of that particular student in the destination page on the right hand side.	10 marks for full implementation and correct navigation, else part marking for correct components